

Laboratory 2 NLP Semantic Similarity 2026

Section A

1. Study examples of Chapter 3 of NLTK online book, provide script that uses beautiful soup to extract text content from 5 bbc news website of your choice. Then extract the tokens of each document using NLTK word tokenizer. For each document (website), the script should display the listing of all tokens in alphabetical order. Display the size of vocabulary of each document.
2. Write a script that filters the tokens of each document by removing uncommon character, numbers, tokens that contains either one or two characters, lower-case the tokens and remove the default NLTK English stopwords. Next, the filtered tokens are normalized using wordnet tokenizer to retrieve the root form of the token. Then display the generated vocabulary of each document in alphabetical order.
3. Use Tf.idf vector tokenizer to restrict the number of vocabulary by selecting appropriate threshold for rare term occurrence and frequency, and then generate the tf-idf vector for each document.
4. Write a script that selects three tokens at random from 1) prior to the filtering process. Assume the Q contains these three tokens through logical AND operator.
5. Suggest a script that uses Boolean retrieval model to calculate the similarity of each document to the query.
6. Repeat the preceding when tf-idf vector model is used.
7. Repeat the preceding when count-vectorizer instead of tf-idf
8. Repeat the preceding when bigram at word level representation was used.
9. We want to use query expansion- For this purpose, use wordnet lexical database to retrieve up to three synonyms for each word of the Query Q in 4). The expanded list of Q, say Q1.
10. Repeat 5-9) to calculate the similarity of the expanded query Q1 with each document.
11. Suggest a script that displays the part-of-speech tag for each token of each document (after filtering and normalization process). We then assume each document represented as a vector or PoS tag for the 45-dimension tag set, so that the value of a given component of this vector corresponds to number of corresponding tags occurring in that document.
12. Calculate the similarity between each pair of documents where the similarity is computed as the cosine similarity value of the vectors in 11)

Section B

1. Study Section 5 of Chapter 2 of NLTK online book, and try to reproduce the coding examples and try to use your own examples of wording to identify the synsets, hyponyms, hypernyms, and various semantic similarity between two words of your choice. Suggest a script that retrieves the first hypernym and the list of all hyponyms of words 'house' and 'train'.
2. Suggest another script that extracts all synonyms and antonyms of "house" and "train" and rank them in the order of their frequencies.
3. Suggest a script that calculates the path similarity between words "house" and "train", and another one computes the Wu and Palmer semantic similarity. Display the hierarchical representation of the synsets and roots highlighting the least common subsumer node. Identify the appropriate synset of each words (house and train) for which the semantic similarity score refers to for each case (path and Wu & Palmer). Note that by

default the semantic similarity between two words corresponds the configuration of synsets for which the highest semantic similarity can be obtained.

4. Write a script that uses wordnet lexical database to retrieve synonyms and antonyms of each of the words “house” and “train”. Then calculate path similarity and Wu and Palmer similarity between each synonym of “house” and each synonym of “train”. Repeat this process for each antonym of each word, and between antonym and synonym for each word. Reproduce these results and conclude on the effect of synonymy and antonymy relationships in wordnet similarity.
5. Write a script that generates the Word2vec embedding vector for each of the words “house” and “train”. Then use the cosine similarity to calculate the semantic similarity between words “house” and “train”.
6. Repeat 5) when using FastText embedding, Glove embedding and BERT embedding.
7. Consider words of the same family such as “house”, “apartment”, “flat”, “hotel”, “room”, “living room”, “toilet”. Write a script that generates path similarity and Wu & Palmer similarity between any pairs of words. Then provide the result in a table.
8. Repeat the precedings when using word2vec embedding, Glove embedding and BERT embedding vector, then cosine similarity.
9. Derive your own conclusion regarding the compatibility between Wordnet semantic similarity and embedding-based